Method overview · from spatial language to continuous 3D execution

帮我看看二楼厨房台面附近的微波炉、二楼客厅沙发旁的茶几和三楼客厅的电视是不是都还在原位，再顺路确认一下二楼卫生间马桶旁还有没有厕纸，先后顺序你自己安排。

Qwen semantic grounding · no prescribed order

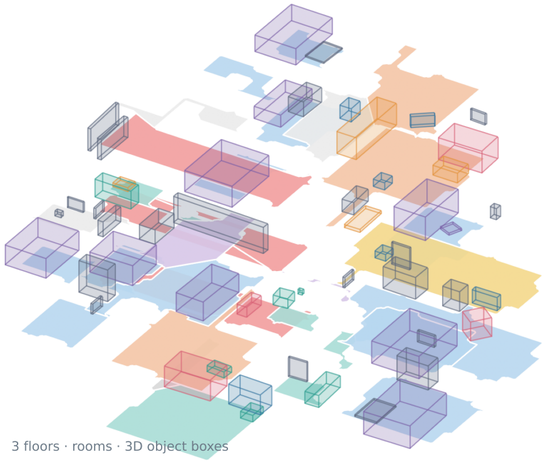
L2 **Microwave**
L2 · kitchen
near countertop

L2 **Coffee table**
L2 · living room
near sofa

L3 **Television**
L3 · living room
instance

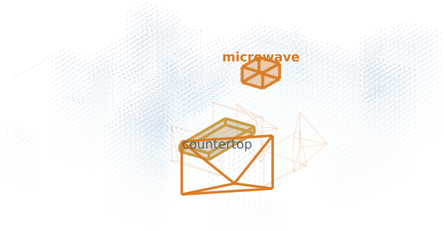
L2 **Toilet paper**
L2 · bathroom
near toilet

① Semantic memory

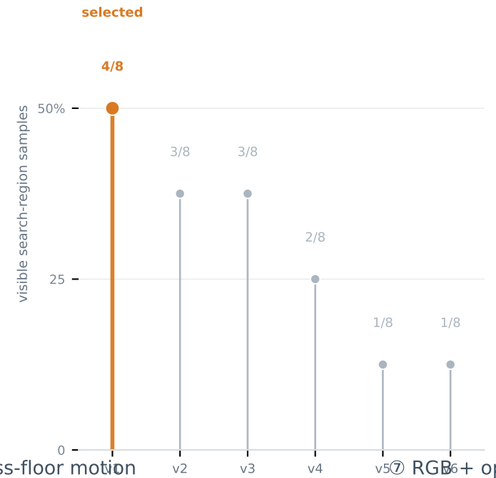


3 floors · rooms · 3D object boxes

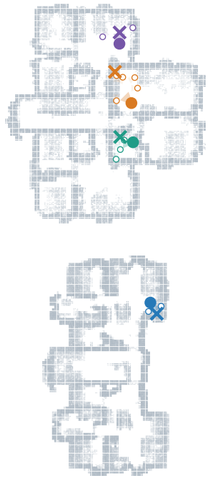
② Spatial grounding



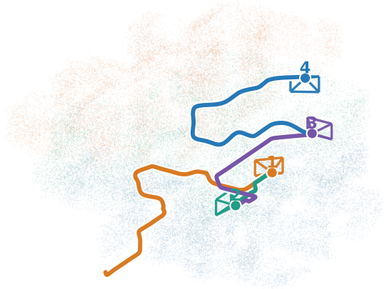
③ Candidate quality



④ Feasible viewpoints · L2 / L3



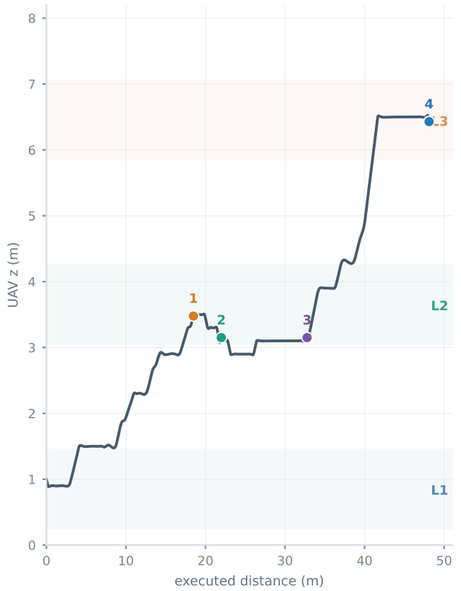
⑤ Joint 3D execution



J = path length + 0.25 flight time + 2 terminal cost

16 feasible views · 48.1 m · 3D A* + B-spline

⑥ Cross-floor motion



⑦ RGB + open-vocabulary







